

Name: _____ Class: _____

Instructions.

Fill in all your answers on this worksheet. You may be asked to hand it up at the end of the lesson. If you are not sure of your answers, ask your classmates first, then only approach your lecturer.

1. Vectors and scalars

A _____ is a quantity that has magnitude.

A _____ is a quantity that has magnitude and direction.

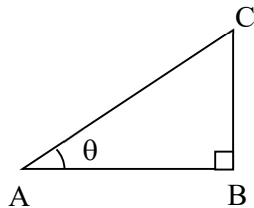
Identify the vectors and scalars in the following list.

Temperature	Height	Velocity	Mass	Acceleration
Distance	Weight	Force	Speed	Time

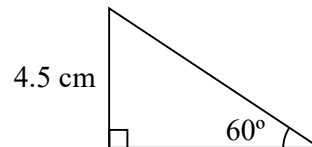
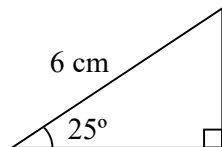
2. Trigonometry.

For the triangle shown, write down all the possible formulae associated with it.

Hint: what does TOA CAH SOH help you to remember?



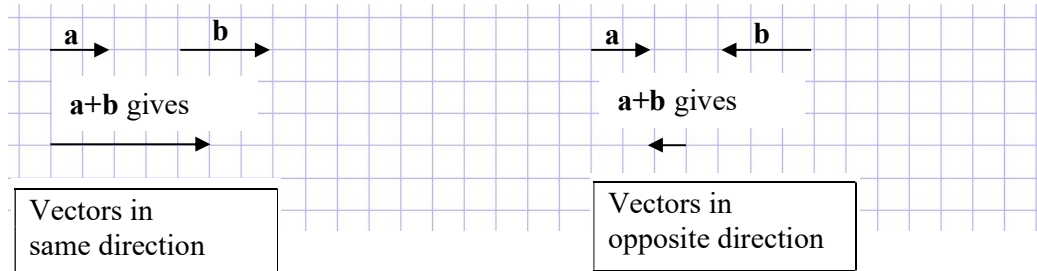
For the following right-angled triangles, determine their unknown lengths.



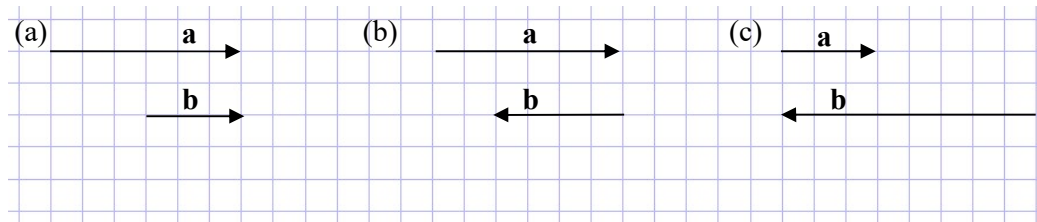
3. Adding vectors (one dimension)

In our course, we take the right/upwards to mean the positive direction, hence our negative direction would be _____.

Note that you can choose your own sign convention too, as long as you apply it consistently in solving the problem.

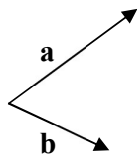


The vectors $\mathbf{a}$ and $\mathbf{b}$ are shown below. In the space below each part, draw the vector sum $\mathbf{a} + \mathbf{b}$.

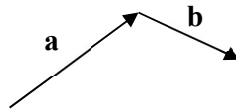


Does $\mathbf{a} + \mathbf{b}$ give you the same result as $\mathbf{b} + \mathbf{a}$?

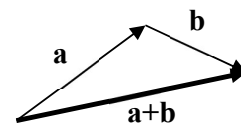
4. Adding vectors (two dimensions)



Problem. You are given two vectors, $\mathbf{a}$ and $\mathbf{b}$. Find $\mathbf{a} + \mathbf{b}$.

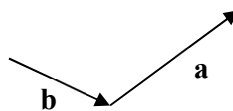


Step 1. Move $\mathbf{b}$ parallel to itself, such that you can join the tail end of vector $\mathbf{b}$ with the head of vector $\mathbf{a}$.

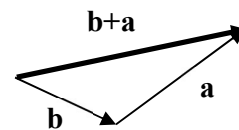


Step 2. $\mathbf{a} + \mathbf{b}$ is obtained by connecting the points as shown.

Now, what if you are asked to find $\mathbf{b} + \mathbf{a}$?



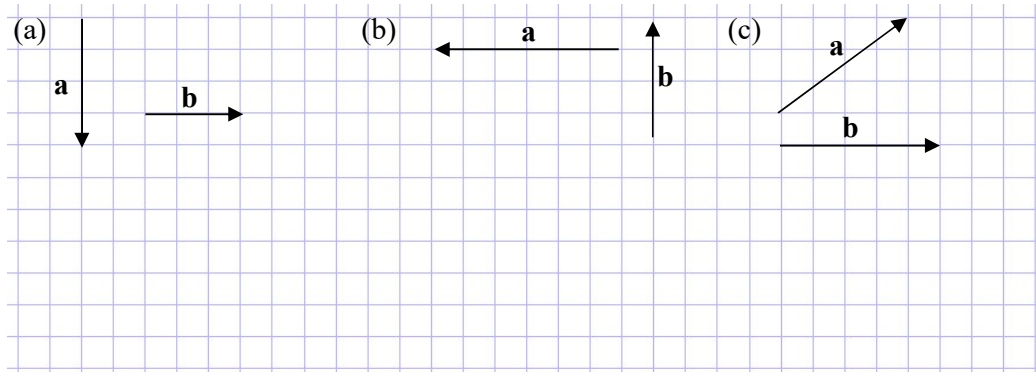
Step 1. Move $\mathbf{a}$ parallel to itself, such that you can join the tail end of vector $\mathbf{a}$ with the head of vector $\mathbf{b}$.



Step 2. $\mathbf{b} + \mathbf{a}$ is obtained by connecting the points as shown.

What do you notice about $\mathbf{b} + \mathbf{a}$ and $\mathbf{a} + \mathbf{b}$?

Section 4 continued. The vectors **a** and **b** are shown below. In the space below each part, draw the vector sum **a+b**.

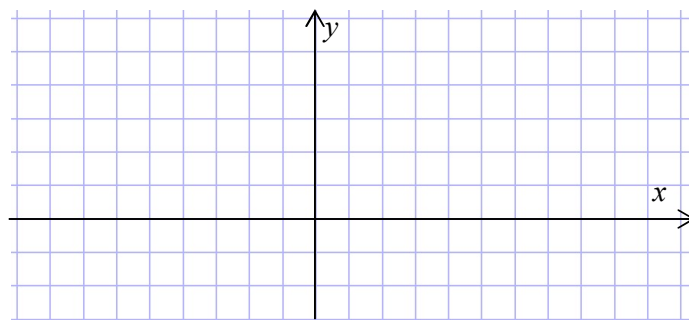


5. Column Vectors.

A column vector describes how a vector is to be drawn. For example, the column vector $\mathbf{p} = \begin{pmatrix} 7 \\ 3 \end{pmatrix}$ tells us that the vector **p** is _____ units long in the *x*-direction and _____ units long in the *y*-direction.

If you draw the column vector starting from the origin, the column vector also tells you the coordinates of the ending point of the vector. For example, the column vector $\mathbf{p} = \begin{pmatrix} 7 \\ 3 \end{pmatrix}$, when drawn from the origin, will end at the point _____.

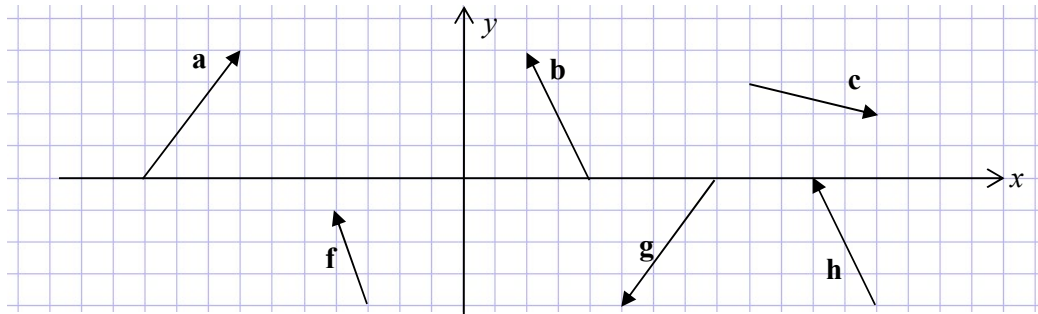
You are given that $\mathbf{a} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$ and $\mathbf{c} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$. On the grid below, draw the vectors **a**, **b** and **c**.



6. Resolving a vector.

The process of expressing vectors into their components is called **resolving a vector**.

In the space below, write the column vector for each of the following vectors, and indicate their x -components and y -components on the diagram.



Column vectors:

a =

b =

c =

f =

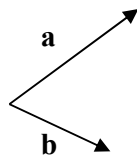
g =

h =

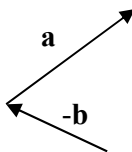
How are vectors **a** and **g** related?

7. Subtracting vectors

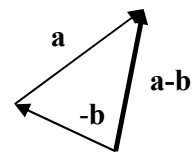
Given a vector **a**, **-a** reverses the direction of the vector, just like how vectors **a** and **g** in Section 6 are related. By the rules of algebra, $\mathbf{a} - \mathbf{b} = \mathbf{a} + (-\mathbf{b})$.



Problem. You are given two vectors, **a** and **b**. Find $\mathbf{a} - \mathbf{b}$.



Step 1. Reverse the direction of vector **b**. You are now ready to perform $\mathbf{a} + (-\mathbf{b})$

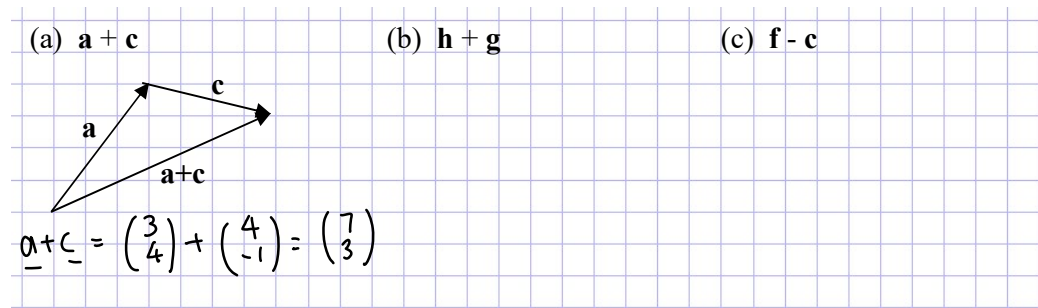


Step 2. $\mathbf{a} - \mathbf{b}$ is obtained by connecting the points as shown.

8. Adding and Subtracting column vectors.

Using the vectors in Section 6,

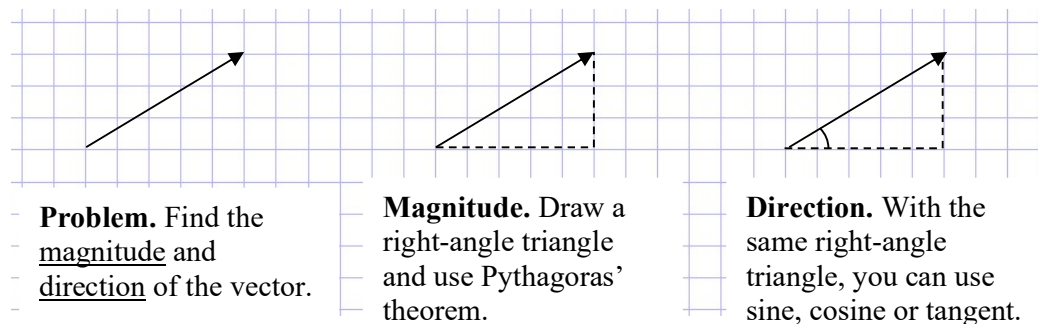
- draw the resultant vectors together with the individual vectors.
- Write the resulting column vector.



9. Magnitude and direction of a vector

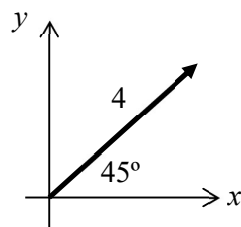
The _____ of a vector is the length of the arrow.

The _____ of a vector is usually specified by taking the angle of the vector with respect to the positive x -axis.

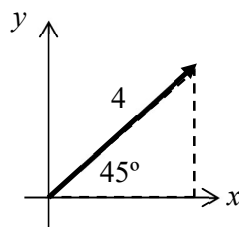


Find the magnitude and direction of vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ in section 6.

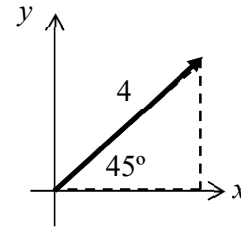
10. Vector components from magnitude and direction.



Problem. Write the column vector given the magnitude and direction.



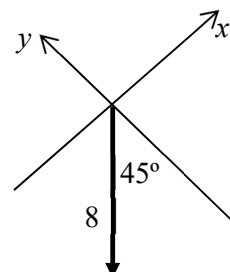
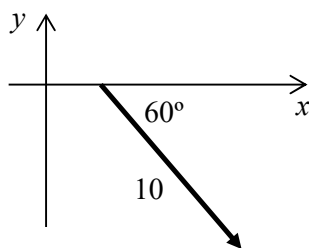
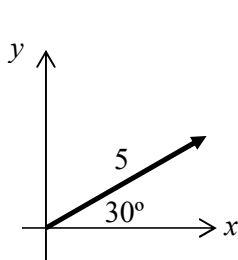
Step 1. Draw a right-angle triangle.



Step 2. You are ready to use sine or cosine to find the length of each side.

- label them on the diagrams below. Be sure to include any negative signs.

b. Write the column vector for each.



11. The x and y components of velocity of a particle are 25 m/s and 60 m/s respectively. Find the magnitude and direction of the velocity vector.

[65 m/s , 67.4° with positive x -axis]

You can always express the answer also with respect to the y -axis as long as you specify. Note that any of the two vectors can be chosen to be along x -axis.

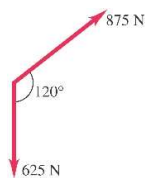
12. Find the resultant of two forces of 10 N and 20 N acting at a point such that the angle between the forces is a) 60° b) 30° . You can measure the angle of the resultant force with respect to the 10 N force.

[26.4 N , 40.9° , 29.1 N , 20.1°]

13. A car changes its velocity from 30 m s^{-1} due East to 25 m s^{-1} due South. Draw a vector diagram to show the initial, final and change of velocity of the car.

14. Use vector components to find the magnitude and direction of the vector needed to balance two vectors shown in figure. Let the 625 N vector be along the $-y$ axis and let the $+x$ axis be perpendicular to it toward the right.

[781 N , 166.1° from $+x$ axis]



15. The game of “Go” is played on a board with 19×19 intersections, as shown in the figure below. Black and white pieces are used in the game.

A child, not knowing the rules of the game, is playing with the Go board. She places one black piece and one white piece on the board and makes them move in the path show.

For each piece, write the vector for the starting position and the ending position. Hence or otherwise, determine the change of displacement vector.

Does the actual path moved by the piece matter?

